

Lemma 3.3. Let $\ell_1, \dots, \ell_j \in \mathbb{R}[\lambda] \setminus \{0\}$ be univariate polynomials of degree ≤ 1 . Also let $D \in \mathbb{Z}$ and $b_1 := (b_{1,1}, \dots, b_{1,j})$ and $b_2 := (b_{2,1}, \dots, b_{2,j}) \in \mathbb{Q}^k$ be vectors whose non zero numerators and denominators divide D . Define $\Psi(\lambda) := (\psi_1(\lambda), \psi_2(\lambda)) := \left(\prod_{i=1}^j \ell_i^{b_{1,i}}(\lambda), \prod_{i=1}^j \ell_i^{b_{2,i}}(\lambda) \right)$. Then, assuming that $\Psi(\mathbb{R})$ is **not** contained in a line, there are no more than $4j^2D^4 + (j+1)D^2 + 3j - 2$ many $\lambda \in \mathbb{C} \setminus \mathbb{R}$ with $\Psi(\lambda) \in \mathbb{R}^2$. PLEASE CORRECT THIS BOUND

Proof of Lemma 3.3: First note that without loss of generality, we can exponentiate the ψ_i and assume that the $b_{i,i'}$ are all integers that are either zero or divide D^2 . Also, note that

- $\Psi(\lambda) = (0, 0) \in \mathbb{R}^2$ for at most j values of λ making some ℓ_i vanish. We will assume henceforth that $\ell_1(\lambda) \cdots \ell_j(\lambda) \neq 0$.
- $\Psi'_1(\lambda) \cdot \Psi'_2(\lambda) = 0$ for at most $2jD^2$ values of λ since both coordinates ψ_1, ψ_2 have isolated zeros since they are non constant.
- $\Psi'_1(\lambda) - \Psi'_2(\lambda) = 0$ for at most jD^2 values of λ , since by our hypothesis about the image of $\mathbb{R}$, the difference $\Psi_1 - \Psi_2$ cannot be a constant function.

So, avoiding at most $j + 3jD^2$ points, we may assume that $\ell_1(\lambda) \cdots \ell_j(\lambda) \neq 0$, $\Psi(\lambda) \neq (0, 0)$, $\Psi'_1(\lambda) \neq 0$, $\Psi'_2(\lambda) \neq 0$ and $\Psi'_1(\lambda) \neq \Psi'_2(\lambda)$.

Next, note that each ψ_i defines a complex analytic function (on the complement of a finite point set in $\mathbb{C}$) and is thus an open mapping. In particular, the inverse images $C_1 := \psi_1^{-1}(\mathbb{R})$ and $C_2 := \psi_2^{-1}(\mathbb{R})$ are finite unions of real analytic curves, and we are trying to compute the cardinality of $I := C_1 \cap C_2 \setminus \mathbb{R}$. To do the latter, let us separate these intersections into **non-transversal** and **transversal**.

In particular, if $\lambda \in I$ is a **non-transversal** intersection, then we must have

$$\psi_1(\lambda) = y_1 \in \mathbb{R}^*, \quad \psi_2(\lambda) = y_2 \in \mathbb{R}^*, \quad \text{and} \quad \left. \frac{d\psi_1^{-1}}{dy} \right|_{y=y_1} = \left. \frac{d\psi_2^{-1}}{dy} \right|_{y=y_2},$$

where the ψ_i^{-1} denote branches of the inverse function of ψ_i satisfying $\psi_i^{-1}(y_1) = \psi_i^{-1}(y_2) = \lambda$ (as well as the right-hand equality above). Via the chain rule, we then have that

$$\frac{1}{\psi'_1(\lambda)} = \frac{1}{\psi'_2(\lambda)}.$$

But this cannot happen since both denominators are different.

To conclude, let us now count the transversal intersections in I . First note that $\Psi(\lambda) \in \mathbb{R}^2$ iff $\psi_1(\lambda) = \bar{\psi}_1(\lambda)$ and $\psi_2(\lambda) = \bar{\psi}_2(\lambda)$, where $(\bar{\cdot})$ denotes complex conjugate. Multiplying through by suitable powers of the ℓ_i , we can then clearly assume that all the exponents in ψ_1 and ψ_2 are nonnegative integers bounded above by $2D^2$. Writing $\lambda = \mu + i\nu$, we can then obtain a 2×2 polynomial system — with **complex** coefficients — of the form $\Phi(\mu, \nu) = (0, 0)$. The real solutions of Φ with $\nu \neq 0$ are exactly the real and imaginary parts of the complex solutions of $(\psi_1 - \bar{\psi}_1, \psi_2 - \bar{\psi}_2)$ we are trying to count. Moreover, the real solutions of $\Phi(\mu, \nu) = (0, 0)$ with $\nu \neq 0$, thanks to our transversality assumption, are isolated points in the **complex** zero set of Φ . The total degree of each equation is bounded above by $2jD^2$, so by Bézout's Theorem, the number of real solutions of Φ is no more than $4j^2D^4$.

We thus obtain that there are no more than $j + (j - 1) + ((j + 1)D^2 + j - 1) + 4j^2D^4 = 4j^2D^4 + (j + 1)D^2 + 3j - 2$ points in I and we are done. ■